

Analyze Data in a Data Warehouse

Certification Track: DP-600 [Fabric Analytics Engineer Associate]

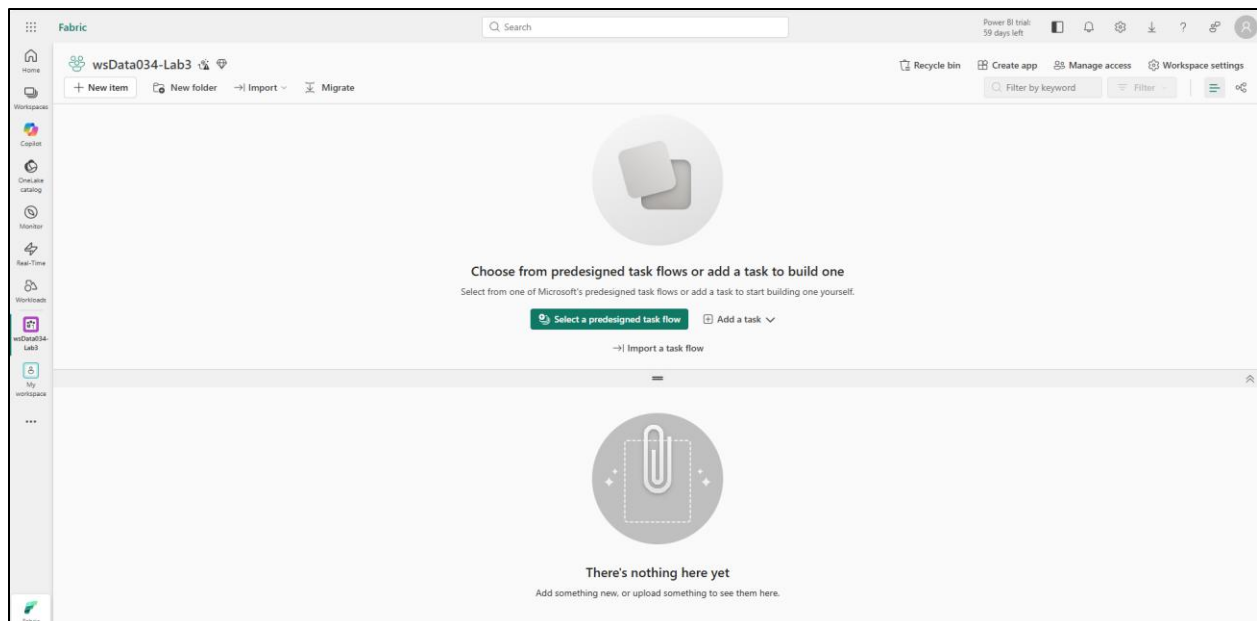
The function of a data warehouse is to move data from source systems, store data that is optimized for analytics, transform data that is ready to use by analytical tools, and analyze data to gain insights for the business.

The purpose of this lab is to showcase the inner workings of a data warehouse by providing a centralized place to store clean, reliable and fast data for analytics to help organizations make data-driven decisions.

1. Create Workspace

This is the starting point for any data project in Fabric. The first step is to create a workspace to hold and organize all your Fabric items.

- Open Microsoft Fabric in your web browser and go to the Fabric home page:
<https://app.fabric.microsoft.com/home?experience=fabric>
- Select Workspace from the left menu.
- Click +New workspace.
- Name your workspace and select Create. Your new workspace should be empty.



2. Create Data Warehouse

To work with large-scale analytics in Microsoft Fabric, a data warehouse offers a full relational database to insert, update, and delete data in tables. These capabilities are not allowed in a Lakehouse's read-only SQL endpoint.

The second step is to create a data warehouse once your new workspace is set up.

- Select ... from the left menu bar and click Create.
- Scroll down to the Data Warehouse section and select Warehouse.
- Name your warehouse and select Create.

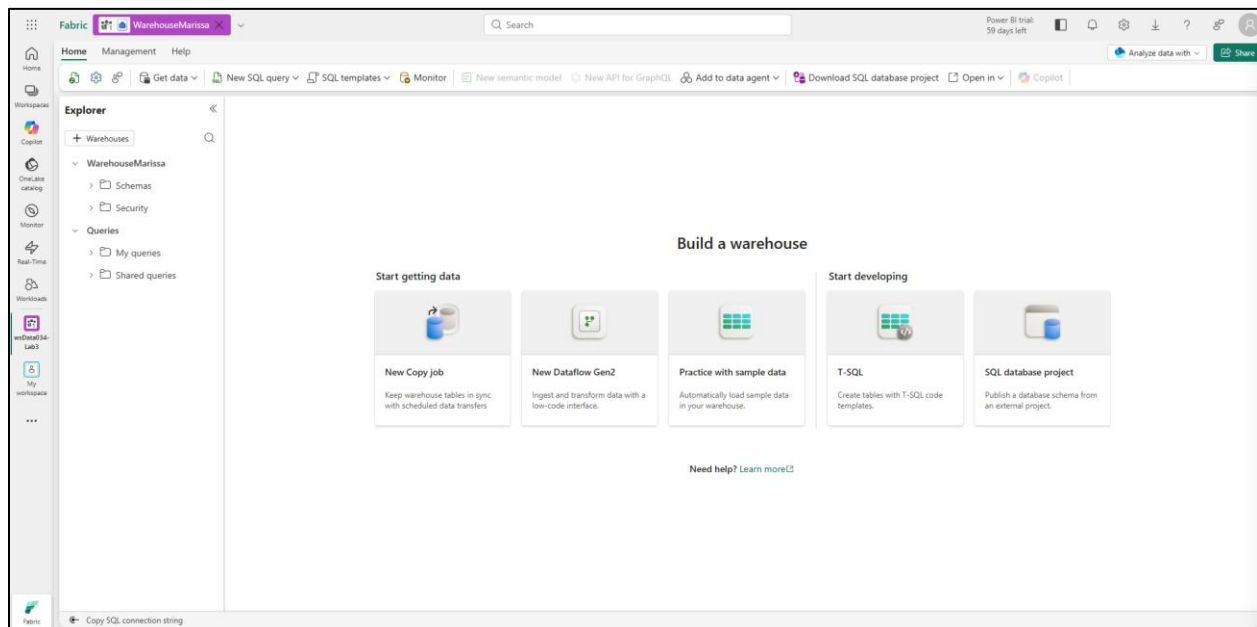
3. Create tables & insert data

A data warehouse in Fabric is a relational database where you can create structured tables and other objects to store your data for analytics. A data warehouse is organized as a star schema, where a fact table relates to dimension tables.

A fact table is the primary source for data analysis, contains numerical data and has many rows. Dimension tables contain descriptive information about the data in the fact table and have fewer rows.

The third step is to create a new table in the data warehouse using T-SQL code and add data to it.

- Now that you have a new warehouse, click on the T-SQL tile where an SQL editor opens up.

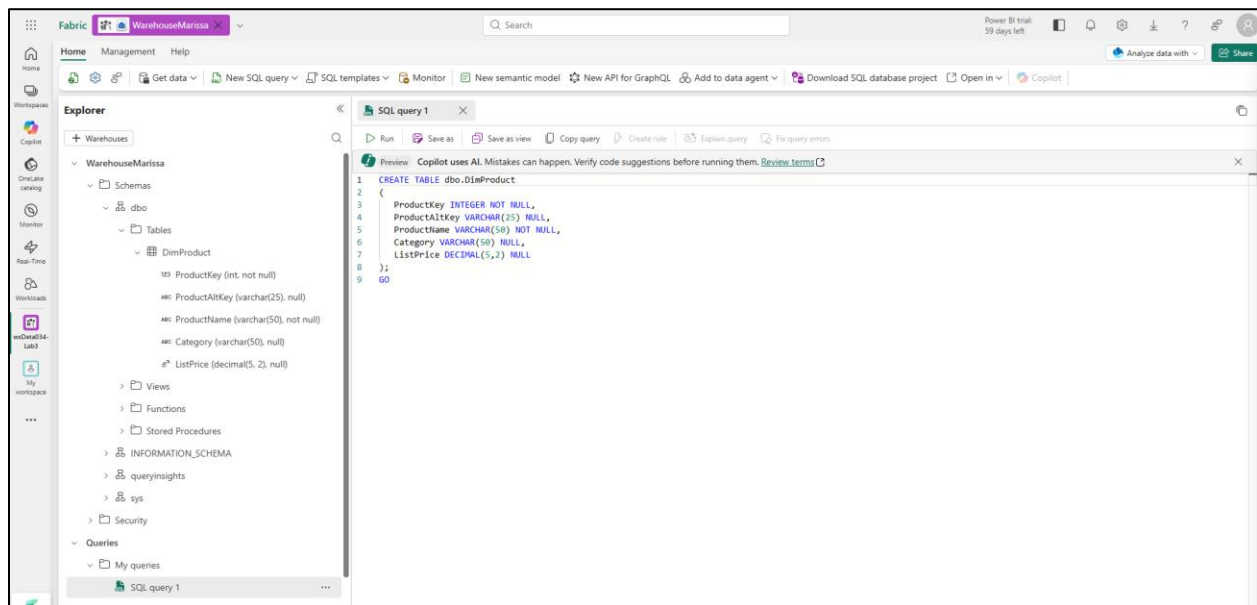


- Copy and paste the following CREATE TABLE statement:

```
CREATE TABLE dbo.DimProduct
(
    ProductKey INTEGER NOT NULL,
    ProductAltKey VARCHAR(25) NULL,
    ProductName VARCHAR(50) NOT NULL,
    Category VARCHAR(50) NULL,
    ListPrice DECIMAL(5,2) NULL
);
GO
```

- Click the ▶ Run button to run the SQL script. A new table is created named DimProduct in the dbo schema of the data warehouse.
- Click the Refresh button on the toolbar under Home to refresh the view.
- In the Explorer pane, expand Schemas-> dbo-> Tables and confirm the DimProduct table appears.

Create table DimProduct:

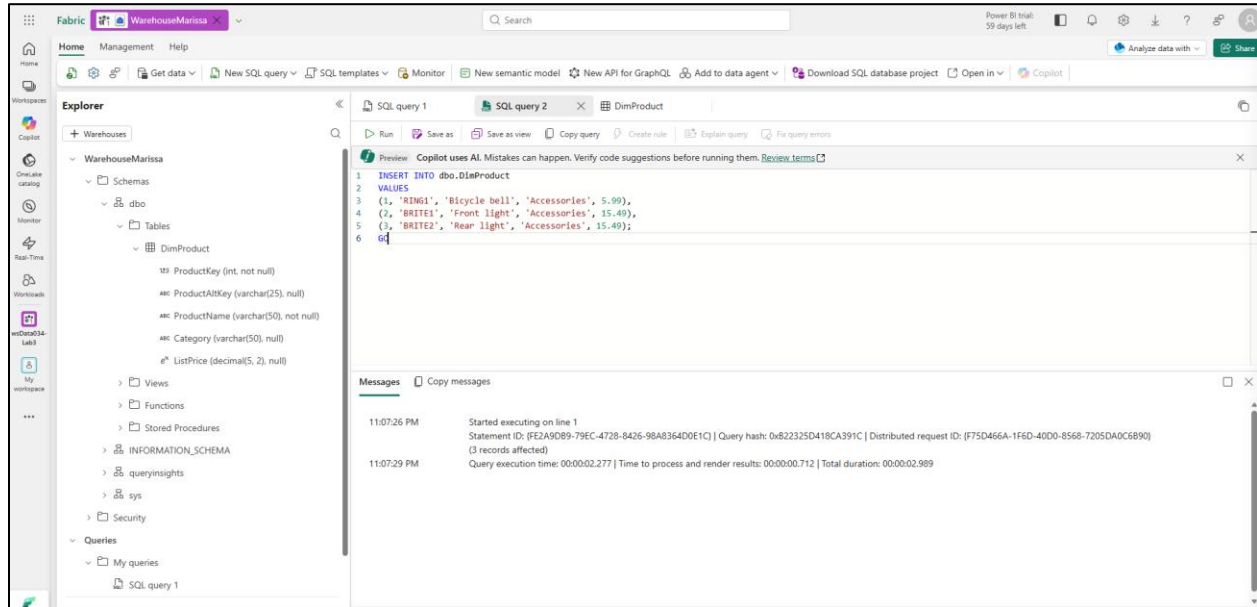


- Click the New SQL Query button on the Home menu tab and create a new query:

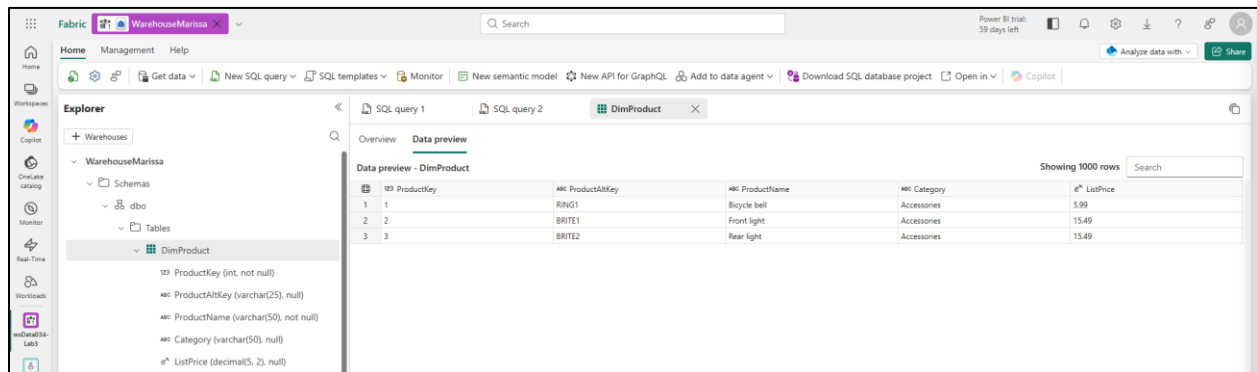
```
INSERT INTO dbo.DimProduct
VALUES
(1, 'RING1', 'Bicycle bell', 'Accessories', 5.99),
(2, 'BRITE1', 'Front light', 'Accessories', 15.49),
(3, 'BRITE2', 'Rear light', 'Accessories', 15.49);
GO
```

- Click the ▶ Run button to run the new query that inserts three rows into the DimProduct table.

Insert data into DimProduct:

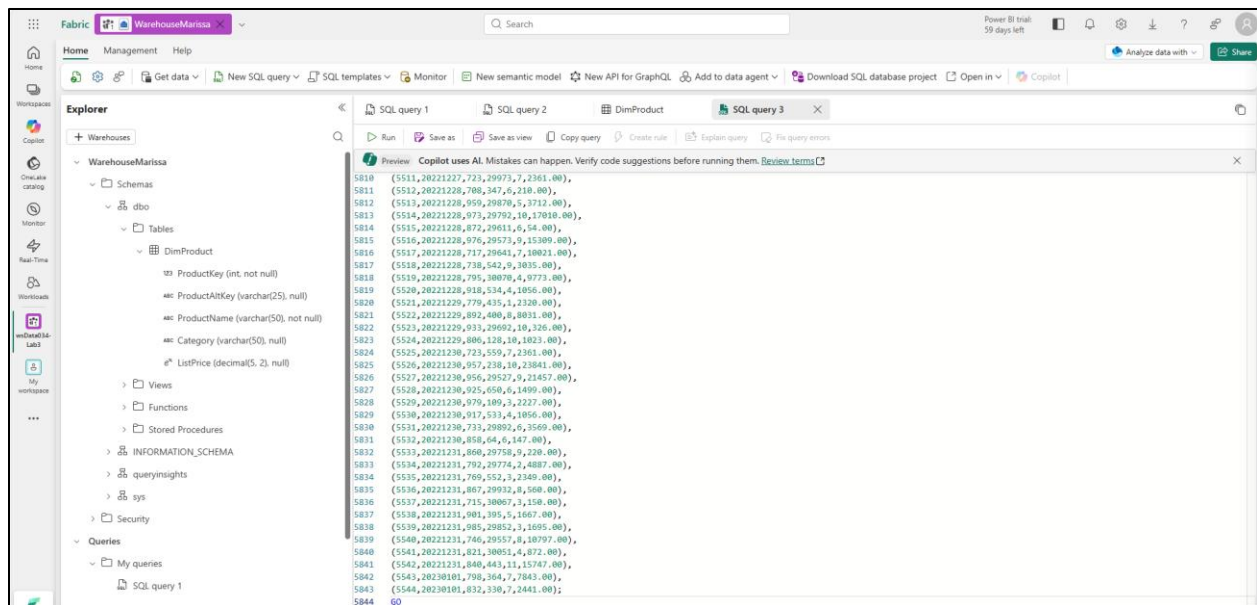


- After it completes, select the DimProduct in the Explorer pane to verify the three rows have been added.



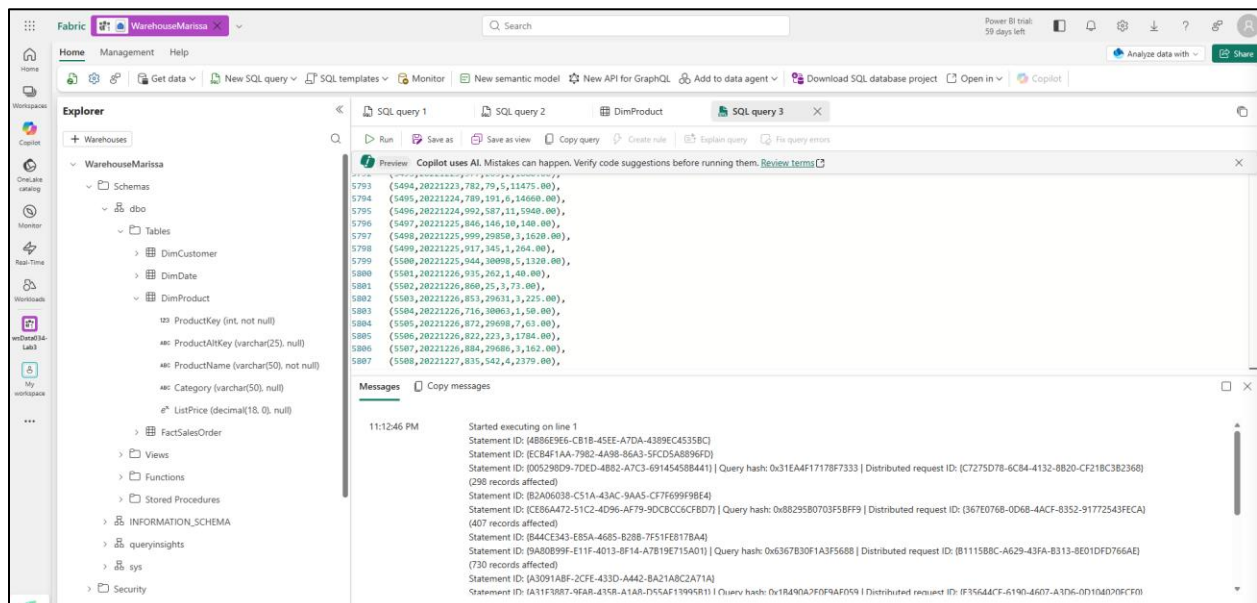
- Next, select New SQL query from the Home tab. Copy and paste the Transact-SQL code from <https://raw.githubusercontent.com/MicrosoftLearning/dp-data/main/create-dw.txt>
- Run the query. The script creates a basic data warehouse schema and loads sample data which typically finishes in 30 seconds.
- Click the Refresh button on the toolbar to update the view.

Simple data warehouse schema with sample data:



- Check in the Explorer pane that the dbo schema in the data warehouse now has four new tables: DimCustomer, DimDate, DimProduct, FactSalesOrder.

dbo schema contains 4 new tables :



4. Query Data Warehouse Tables

A data warehouse is optimized for analytics and centers on grouping and summarizing data. The tables in a relational data warehouse use T-SQL queries to aggregate data with functions like GROUP BY and JOINS.

- Click the New SQL Query button on the Home menu tab to create a new query and run the following code:

```
SELECT d.[Year] AS CalendarYear,
       d.[Month] AS MonthOfYear,
       d.MonthName AS MonthName,
       SUM(so.SalesTotal) AS SalesRevenue
FROM FactSalesOrder AS so
JOIN DimDate AS d ON so.SalesOrderDateKey = d.DateKey
GROUP BY d.[Year], d.[Month], d.MonthName
ORDER BY CalendarYear, MonthOfYear;
```

The screenshot shows the Microsoft Fabric SQL query editor interface. The left sidebar displays the 'Explorer' pane with a tree view of the 'WarehouseMarissa' database, including schemas, tables, and views. The main editor area shows a SQL query (SQL query 4) with the following code:

```
1 SELECT d.[Year] AS CalendarYear,
2       d.[Month] AS MonthOfYear,
3       d.MonthName AS MonthName,
4       SUM(so.SalesTotal) AS SalesRevenue
5 FROM FactSalesOrder AS so
6 JOIN DimDate AS d ON so.SalesOrderDateKey = d.DateKey
7 GROUP BY d.[Year], d.[Month], d.MonthName
8 ORDER BY CalendarYear, MonthOfYear;
```

Below the query editor, the 'Results' pane displays the output of the query as a table with 4 columns: CalendarYear, MonthOfYear, MonthName, and SalesRevenue. The table contains 12 rows of data, representing the months of the year 2021.

CalendarYear	MonthOfYear	MonthName	SalesRevenue
2021	1	January	687029
2021	2	February	836937
2021	3	March	869863
2021	4	April	916273
2021	5	May	961241
2021	6	June	659677
2021	7	July	572189
2021	8	August	794832
2021	9	September	929086
2021	10	October	614116
2021	11	November	890905
2021	12	December	820810

- Modify the query to add a second dimension to the aggregation:

```
SELECT d.[Year] AS CalendarYear,
       d.[Month] AS MonthOfYear,
       d.MonthName AS MonthName,
       c.CountryRegion AS SalesRegion,
       SUM(so.SalesTotal) AS SalesRevenue
FROM FactSalesOrder AS so
JOIN DimDate AS d ON so.SalesOrderDateKey = d.DateKey
JOIN DimCustomer AS c ON so.CustomerKey = c.CustomerKey
GROUP BY d.[Year], d.[Month], d.MonthName, c.CountryRegion
ORDER BY CalendarYear, MonthOfYear, SalesRegion;
```

- Run the code. The results now include sales revenue aggregated by year, month, and sales region.

The screenshot shows the Microsoft Fabric interface with a workspace named 'WarehouseMarissa'. In the 'SQL query 4' tab, the following SQL query is executed:

```

1 SELECT d.[Year] AS CalendarYear,
2        d.[Month] AS MonthOfYear,
3        d.MonthName AS MonthName,
4        c.CountryRegion AS SalesRegion,
5        SUM(so.SalesTotal) AS SalesRevenue
6 FROM FactSalesOrder AS so
7 JOIN DimDate AS d ON so.SalesOrderDateKey = d.DateKey
8 JOIN DimCustomer AS c ON so.CustomerKey = c.CustomerKey
9 GROUP BY d.[Year], d.[Month], d.MonthName, c.CountryRegion
10 ORDER BY CalendarYear, MonthOfYear, SalesRegion;

```

The results are displayed in a table with the following columns: CalendarYear, MonthOfYear, MonthName, SalesRegion, and SalesRevenue. The data shows sales revenue for each month of 2021, aggregated by region.

CalendarYear	MonthOfYear	MonthName	SalesRegion	SalesRevenue
2021	1	January	Canada	78554
2021	1	January	United Kingdom	31025
2021	1	January	United States	226684
2021	2	February	Canada	51482
2021	2	February	United Kingdom	17950
2021	2	February	United States	341371
2021	3	March	Canada	126748
2021	3	March	United Kingdom	18652
2021	3	March	United States	205335
2021	4	April	Canada	142429
2021	4	April	United Kingdom	24807
2021	4	April	United States	345539

5. Create a View

A data warehouse is similar to a relational database in that it supports SQL features like creating views and stored procedures. A view is a virtual table that displays the results of SQL queries from one or more underlying tables but does not physically store any data. A view allows you to reuse logic without duplicating it.

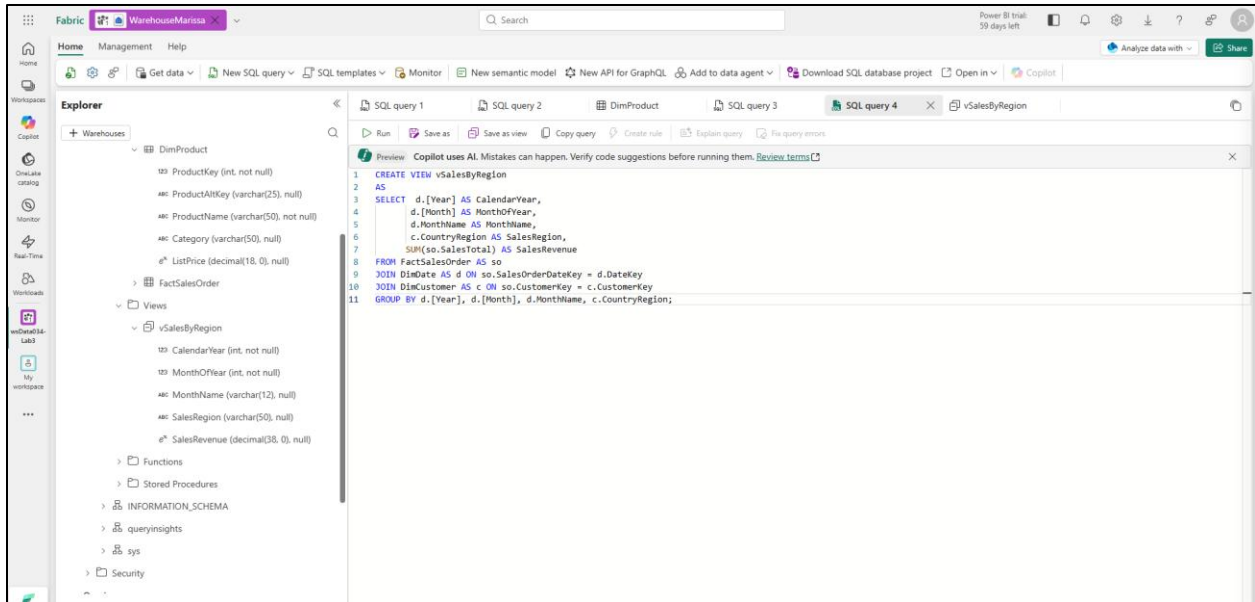
- Modify the previous query and replace with the following code:

```

CREATE VIEW vSalesByRegion
AS
SELECT d.[Year] AS CalendarYear,
       d.[Month] AS MonthOfYear,
       d.MonthName AS MonthName,
       c.CountryRegion AS SalesRegion,
       SUM(so.SalesTotal) AS SalesRevenue
FROM FactSalesOrder AS so
JOIN DimDate AS d ON so.SalesOrderDateKey = d.DateKey
JOIN DimCustomer AS c ON so.CustomerKey = c.CustomerKey
GROUP BY d.[Year], d.[Month], d.MonthName, c.CountryRegion;

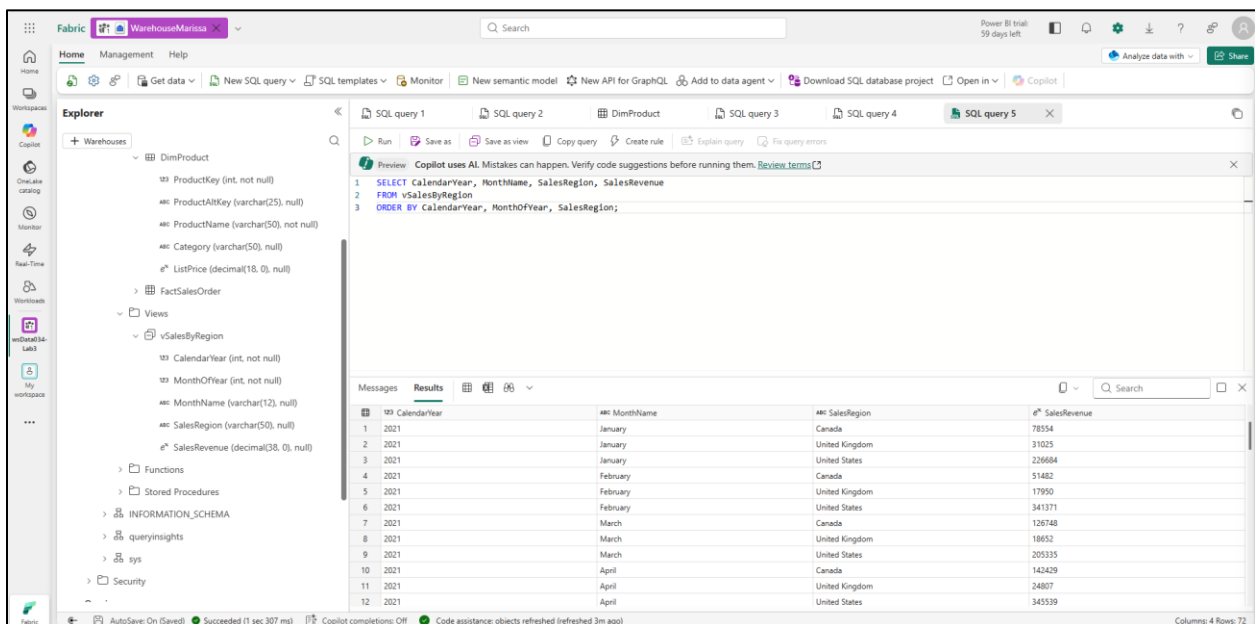
```

- Run the query to create the view.
- Click the Refresh button on the toolbar to update the view and verify the new view appears in the Explorer pane.



- Create a new query below:

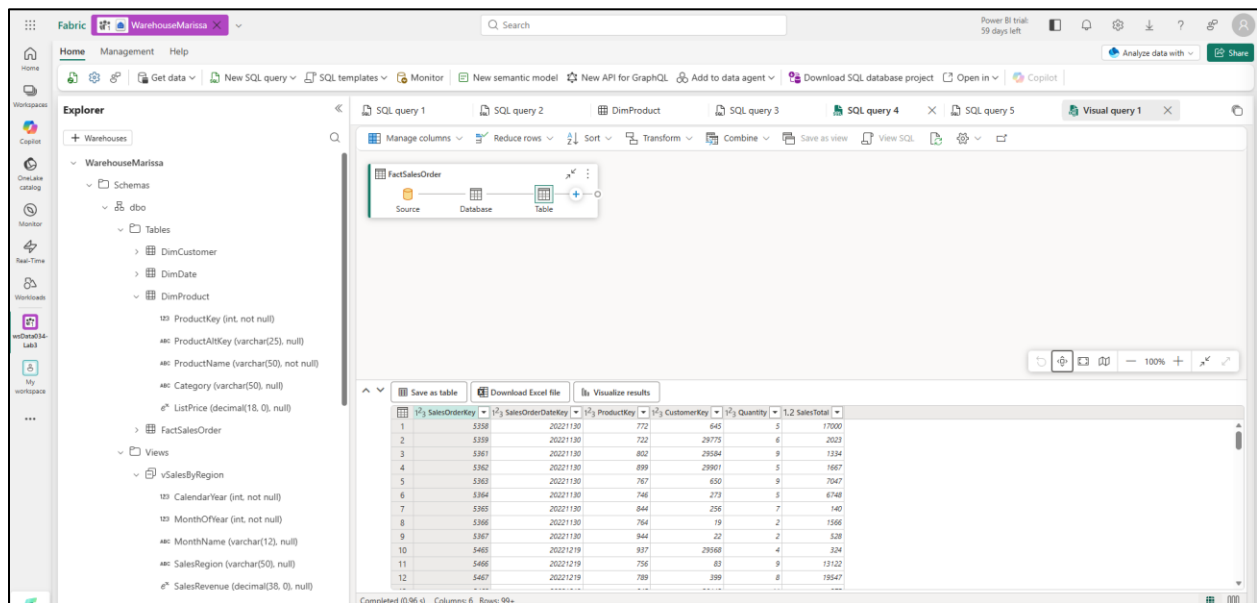
```
SELECT CalendarYear, MonthName, SalesRegion, SalesRevenue
FROM vSalesByRegion
ORDER BY CalendarYear, MonthOfYear, SalesRegion;
```



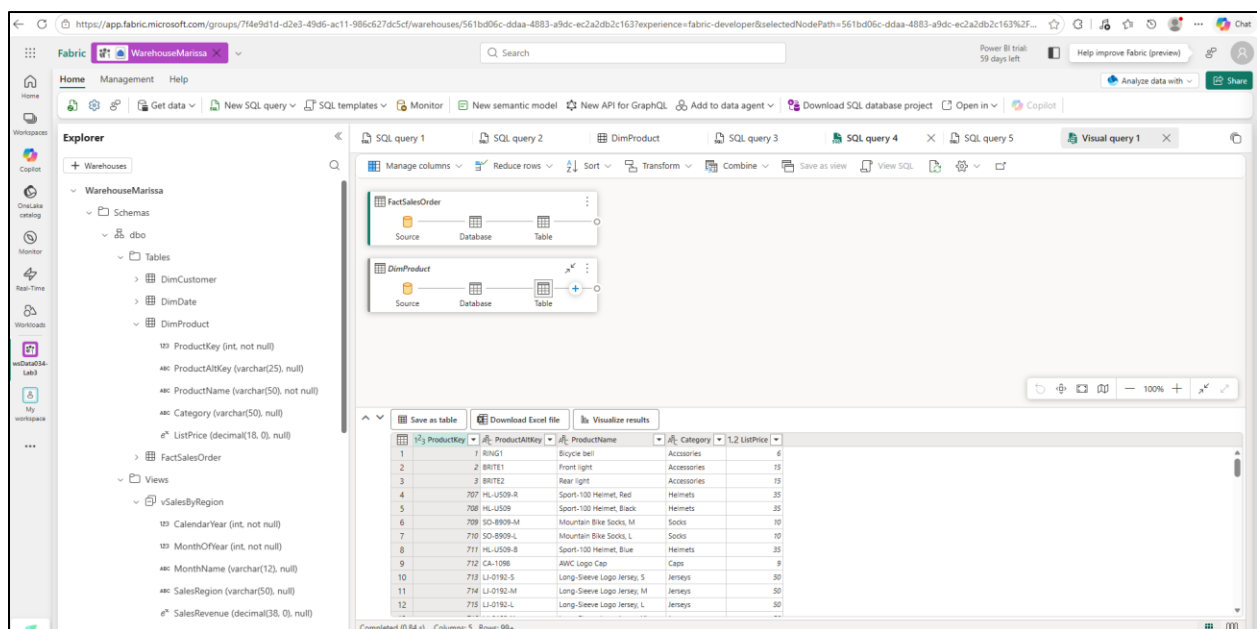
6. Create a Visual Query

You can build queries without writing SQL code by using the graphical query designer with easy point and click steps similar to Power Query Online. For more advanced transformations, you can use Power Query's M (Mashup) language.

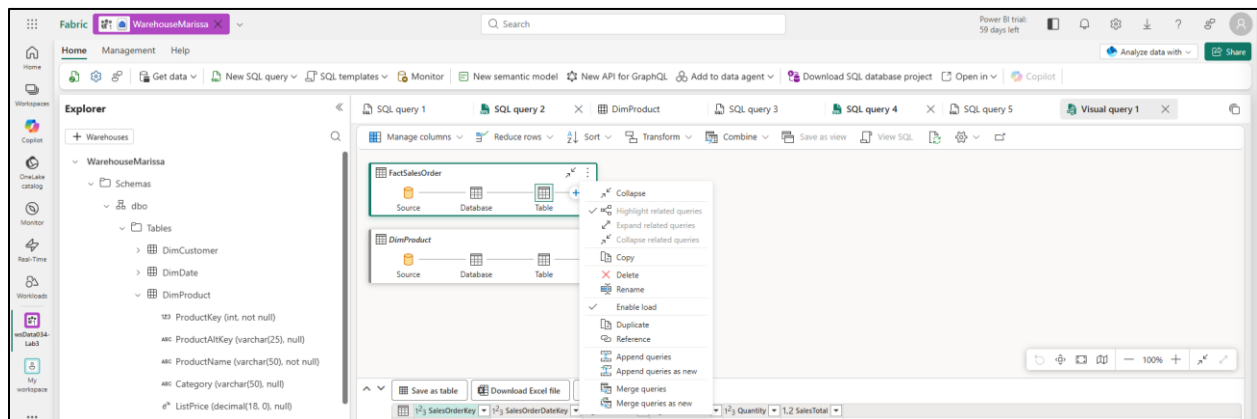
- Create a new visual query: On the Home menu, open New SQL query and select New visual query.



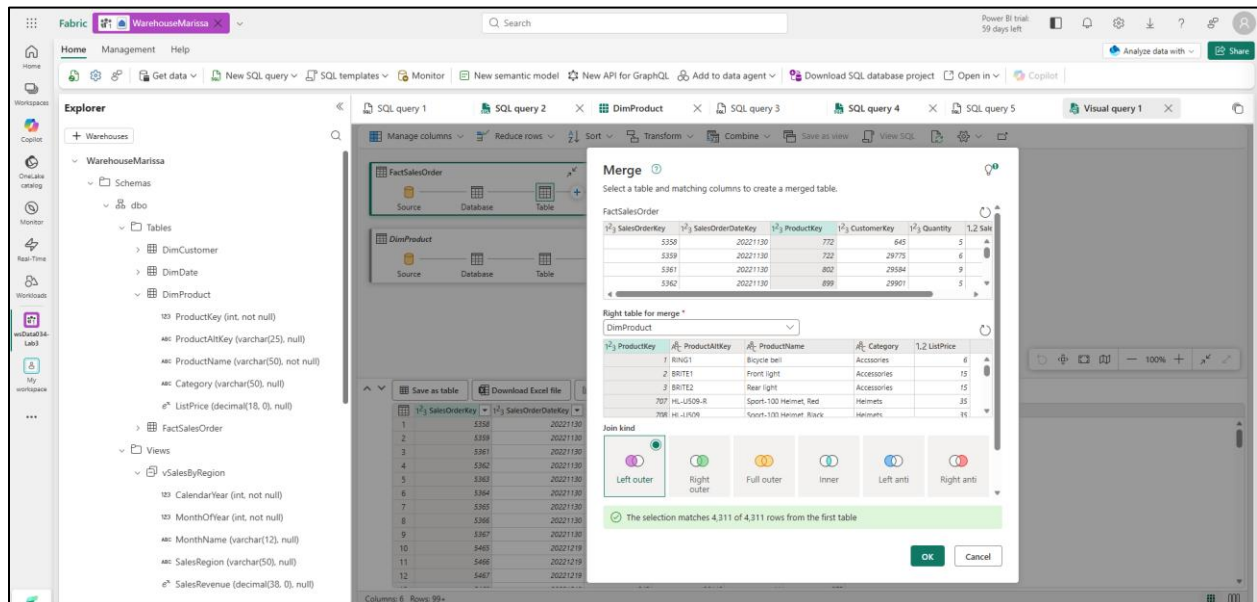
- Add your tables: Drag FactSalesOrder onto the canvas. Drag DimProduct onto the canvas. Both tables are now visible.



- Merge the tables: On the FactSalesOrder table, select the (+) button.



- In the merge window:
 - Choose DimProduct as the right table for merge.
 - Select ProductKey in both tables.
 - Keep the default to Left outer join as the join kind.
 - Click OK.



- Expand the merged column:
 - In the preview pane, you will see a new DimProduct column added to FactSalesOrder.
 - Expand it using the arrow beside the column name.
 - Check off ProductName and click OK.

Fabric WarehouseMarissa

Home Management Help

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Explorer

Warehouses

WarehouseMarissa

Schemas

dbo

Tables

DimCustomer

DimDate

DimProduct

ProductKey (int, not null)

ProductAltKey (varchar(25), null)

ProductName (varchar(50), not null)

Category (varchar(50), null)

ListPrice (decimal(18, 0), null)

FactSalesOrder

Views

vSalesByRegion

CalendarYear (int, not null)

MonthOfYear (int, not null)

MonthName (varchar(12), null)

SalesRegion (varchar(50), null)

SalesRevenue (decimal(38, 0), null)

SQL query 1 SQL query 2 DimProduct SQL query 3 SQL query 4 SQL query 5 Visual query 1

Manage columns Reduce rows Sort Transform Combine Save as view View SQL

DimProduct

Source Database Table

FactSalesOrder

Source Database Table Merged queries

Save as table Download Excel file Visualize results

	SalesOrderKey	SalesOrderDateKey	ProductKey	CustomerKey	Quantity	SalesTotal	DimProduct
1	5358	20221130	772	645	5	17000	Table
2	5359	20221130	722	29775	6	2023	Table
3	5361	20221130	802	29584	9	1334	Table
4	5362	20221130	899	29901	5	1667	Table
5	5363	20221130	767	650	9	7047	Table
6	5364	20221130	746	273	5	6748	Table
7	5365	20221130	844	256	7	140	Table
8	5366	20221130	754	19	2	1568	Table
9	5367	20221130	844	22	2	539	Table
10	5465	20221219	937	29568	4	324	Table
11	5466	20221219	756	83	9	13122	Table
12	5467	20221219	789	399	8	19547	Table

Completed (1.17 s) Columns: 7 Rows: 12

Fabric WarehouseMarissa

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Explorer

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WarehouseMarissa

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DimProduct

ProductKey (int, not null)

ProductAltKey (varchar(25), null)

ProductName (varchar(50), not null)

Category (varchar(50), null)

ListPrice (decimal(18, 0), null)

FactSalesOrder

Views

vSalesByRegion

SQL query 1 SQL query 2 DimProduct SQL query 3 SQL query 4 SQL query 5 Visual query 1

Manage columns Reduce rows Sort Transform Combine Save as view View SQL

DimProduct

Source Database Table

FactSalesOrder

Source Database Table Merged queries

Search

(Select all)

ProductKey

ProductAltKey

ProductName

Category

ListPrice

Use original column name as prefix

OK Cancel

	SalesOrderKey	SalesOrderDateKey	ProductKey	CustomerKey	Quantity	SalesTotal	DimProduct
1	5358	20221130	772	645	5	17000	Table
2	5359	20221130	722	29775	6	2023	Table
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4	5362	20221130	899	29901	5	1667	Table
5	5363	20221130	767	650	9	7047	Table

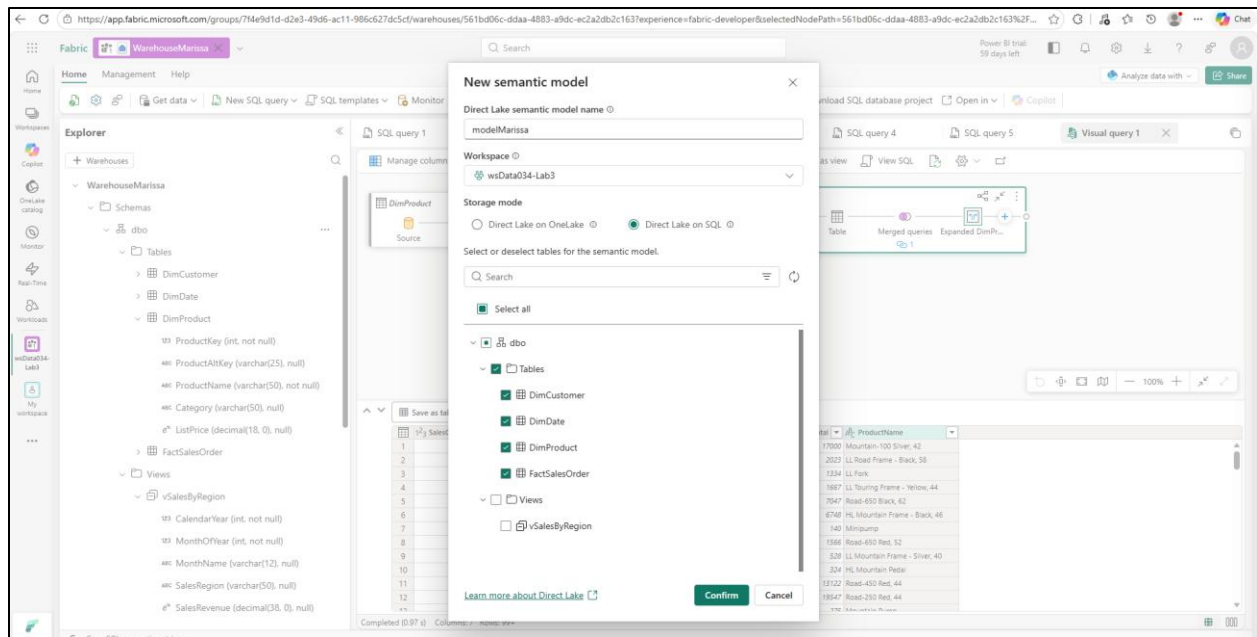
- You can now use the ProductName column to filter the data in the query.
- At this point you can review the output of the query by choosing Visual results or Download Excel file.

The screenshot displays the Microsoft Fabric interface. On the left, the 'Explorer' pane shows a workspace named 'WarehouseMarissa' with a schema 'dbo' containing several tables, including 'DimProduct' and 'FactSalesOrder'. The 'DimProduct' table is expanded, showing columns like 'ProductKey', 'ProductAltKey', 'ProductName', 'Category', and 'ListPrice'. The main area shows a data model diagram with 'DimProduct' as a source table connected to 'FactSalesOrder'. Below the diagram, the 'Visualize results' tab is active, displaying a table with 12 columns: 'SalesOrderKey', 'SalesOrderDateKey', 'ProductKey', 'CustomerKey', 'Quantity', 'SalesTotal', and 'ProductName'. The table contains 12 rows of data, including product names like 'Mountain-100 Silver, 42', 'Road Frame - Black, 58', and 'Touring Frame - Yellow, 44'. The bottom status bar indicates 'Completed (0.97 s) Columns: 7 Rows: 99'.

7. Define a Data Model (Optional)

A data model defines the structure of the whole system and how tables relate. This helps to organize information to ensure reports, dashboards and analysis are consistent, accurate, and aligned across a business.

The Data Modeling feature can only be used if you have a Power BI license or Fabric F64 SKU. Students do not have access to this feature therefore cannot complete this step.



8. Clean up Resources

It is important to delete the resources after completing the lab to stop unnecessary capacity consumption which incur costs.

- Select the icon for your workspace on the left to view all items in the workspace.
- In the toolbar, select Workspace settings.
- In the General section, select Remove this workspace.

9. Reflection

In this lab, I learned that a data warehouse is used for analytics and the end result stems from having clean, structured and reliable data that's prepared before any report or dashboard is created. A data warehouse takes advantage of a relational database's capabilities by utilizing structured tables, defined schemas, and full T-SQL operations for transforming data. The key is to be able to run complex queries at high speed across large volumes of structured data, producing consistent, reliable analytical results. The process starts with creating a Fabric workspace, building a data warehouse, and defining and populating relational tables using T-SQL. Then you execute analytical queries with joins, filters and aggregations. You can also create views and visual queries. All these steps are critical in transforming raw data into meaningful insights for the business.